🔑 **Answer Key — Teacher Copy · Lesson 5-9**

MSTAR PRACTICE WORKSHEET

Lesson 5-9: Area of Regular Polygons

Grade 6 Mathematics · Standard 6.GR.1

These are MSTAR-style questions written for this curriculum to rehearse the format. They are not official Maryland assessment items. MSTAR — Maryland's new state test, first given in spring 2027 — uses the same kinds of questions you see here: selected response, select-all, two-part evidence questions, and written responses.

Part A — Skills Check

One point each.

Question 1 SELECTED RESPONSE — CORRECT: B

A regular hexagon is divided into 6 equal triangles from the center. Each triangle has a base of 6 ft and a height of 5.2 ft. What is the area of one triangle? Use the given heights; nonsquare heights are rounded, so their areas are estimates.

A. 11.2 sq ft

B. 15.6 sq ft

C. 15.6 ft

D. 31.2 sq ft

A = ½ × b × h = ½ × 6 × 5.2 = 15.6 square feet.

Question 2 SELECTED RESPONSE — CORRECT: C

A regular hexagonal skylight has 6 central triangles, each with base 6 ft and rounded height 5.2 ft. Using those measurements, what is the approximate total area?

A. 15.6 sq ft

B. 62.4 sq ft

C. 93.6 sq ft

D. 187.2 sq ft

Total area = 6 triangles × 15.6 sq ft = 93.6 square feet.

Question 3 SELECTED RESPONSE — CORRECT: C

A regular hexagon is split into 6 triangles, each with base 4 cm and height 3.5 cm. What is the total area? Use the given heights; nonsquare heights are rounded, so their areas are estimates.

A 14 sq cm

B 21 sq cm

C 42 sq cm

D 84 sq cm

Each triangle: ½ × 4 × 3.5 = 7 sq cm. Total: 6 × 7 = 42 sq cm.

Question 4 SELECTED RESPONSE — CORRECT: B

How many triangles can you make from the center of a regular pentagon?

A 3

B 5

C 6

D 10

A regular pentagon has 5 sides, so drawing lines from the center to each vertex creates 5 equal triangles.

Question 5 SELECTED RESPONSE — CORRECT: D

A regular hexagonal floor tile splits from its center into 6 congruent triangles, each with an area of 12 cm². What is the total area of the tile?

A 12 cm²

B 18 cm²

C 36 cm²

D 72 cm²

A hexagon has 6 sides, so it splits into 6 equal triangles. Total = 6 × 12 = 72 cm².

Question 6 SELECTED RESPONSE — CORRECT: C

A regular pentagonal tile splits into 5 congruent triangles. Each triangle has a base of 4 in and a height of about 2.8 in. What is the total area of the tile? Use the given heights; nonsquare heights are rounded, so their areas are estimates.

A 5.6 in²

B 11.2 in²

C 28 in²

D 56 in²

One triangle = ½ × 4 × 2.8 = 5.6 in². A pentagon has 5 triangles, so total = 5 × 5.6 = 28 in².

Part B — Test-Format Questions

Scoring notes and distractor rationales below each item.

Question 7 TWO-PART QUESTION

Part A — correct answer: C

A regular hexagon is decomposed from its center into 6 identical triangles. Each triangle has a base of 8 cm and a height of 7 cm. What is the area of the hexagon? Use the given rounded height to estimate the area.

A. 28 square centimeters

B. 56 square centimeters

C. 168 square centimeters

D. 336 square centimeters

Each triangle = ½ × 8 × 7 = 28 square centimeters. Six of them: 6 × 28 = 168 square centimeters.

- **A:** That is ONE triangle. The hexagon is made of six of them.
- **B:** That is two triangles. All six make up the hexagon.
- **D:** The triangles were never halved. Each is half of 8 times 7, which is 28, and six of those give 168.

Part B — correct answer: C

Why can a regular polygon be decomposed into identical triangles from its center?

A. Every polygon can be split into identical triangles.

B. Triangles are the only shape with a known area formula.

C. All its sides are equal and all its angles are equal, so each center-to-side triangle is congruent.

D. The center of a polygon is always a vertex.

Regular means equal sides and equal angles, so the triangles formed from the center to each side are congruent and share the same height.

Question 8 SELECT ALL THAT APPLY — CORRECT: A, B, D, E

Select ALL statements that are true about the area of a regular polygon:

A. It equals the number of sides times the area of one center triangle.

B. Each center triangle has the same height for a regular polygon.

C. Only regular polygons with an even number of sides can be decomposed this way.

D. A regular polygon has all sides equal and all angles equal.

E. The height of each center triangle is measured perpendicular to a side.

A, B, D, and E are true. C is false — a regular pentagon has 5 sides and decomposes into 5 congruent triangles just as well.

Part C — Show Your Thinking

Model answer and scoring guidance.

Question 9 WRITTEN RESPONSE · 2 POINTS

A regular hexagon and a regular pentagon are each divided into congruent central triangles. In both polygons, each central triangle has area 15 sq ft. Which polygon has greater total area? Explain why the triangle count matters. The two polygons have different side lengths.

Model answer: The hexagon has area 6 × 15 = 90 sq ft. The pentagon has area 5 × 15 = 75 sq ft. The hexagon has 15 sq ft more because it has one additional triangle of that area. Equal triangle areas do not require equal bases and heights.

Award 2 points for a complete explanation with correct mathematics, 1 point for a correct answer with thin reasoning, 0 for an unrelated response.

Answer Key. Score written responses with the rubric and guidance above; award partial credit per the 1-point rows.